BioSpecInfo

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

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Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	`node --check` on the critical JavaScript blocks on every change.	✅ In use
End-to-end (E2E) testing	Automation with **Playwright** on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: **0 non-network errors**).	✅ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✅ In use
Manual cross-device testing	Testing on real devices (see §4).	✅ In use
Automated unit-test suite (Vitest/Jest)	—	🔜 Roadmap
Coverage ≥ 90%	—	🔜 Roadmap

> Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ^{1}H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	**SDBS** (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	**0.31 ppm**
Signals within ±0.5 ppm	**89%**

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	**NIST Chemistry WebBook**
Mean Absolute Error (MAE)	**11 cm^{-1}**
Bands within ±30 cm^{-1}	**90%**

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol → $C_{27}H_{46}O$, 4 rings, 8 stereocentres*.

> **Benchmark data source:** measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

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4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: **≈ 4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do

not fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise

functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS

pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases

(NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

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6. Release acceptance criteria

- ✅ `node --check` passes without errors on the modified modules.
- ✅ Playwright E2E: 0 non-network JS errors.
- ✅ No visible regression in the touched sections.
- ✅ Service Worker cache version bumped.
- ✅ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
- ✅ `verifica_guida`: whatever the documentation promises exists in the code.

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7. Verification battery — state at version `bsi-v217`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	`verifica-farmaci`, `test_spettri`, `test_assi`, `test_assi_canvas`, `test_costanti`, `audit_dati`	structure vs molecular weight, functional-group recognition, axis conventions, physical constants

Family	Benches	Subject
AI agent	`test_ko`, `test_404`, `test_503`, `test_firma`, `test_attesa`, `test_attesalunga`, `test_tetto`, `test_nucleo`, `browser_ko`, `browser_prova`	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	`audit_stabilita`, `audit_promesse`, `audit_quota`, `test_filemanager`, `test_doppioinvio`, `test_sw`, `test_visore3d`	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	`browser_reset`, `browser_proxy`, `browser_proxyui`, `browser_rdkit`, `browser_lab`, `browser_frontiera`, `test_aggiorna`, `test_guidaproxy`	panels, proxy, RDKit laboratory, updates
Consistency	`verifica_guida`	documentation ⟷ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	96 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, **+0 on the four that follow**
Survival of exhausted storage	`localStorage.setItem` forced to throw	10 pages out of 10 stay operational
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in **3,507 ms** (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to **1**

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the

Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.

- Probing `localStorage` exhaustion with a one-byte write proves nothing: after

saturation, one byte always finds room.

- The ticks of an axis drawn on `<canvas>` are not readable as text: they must

be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.